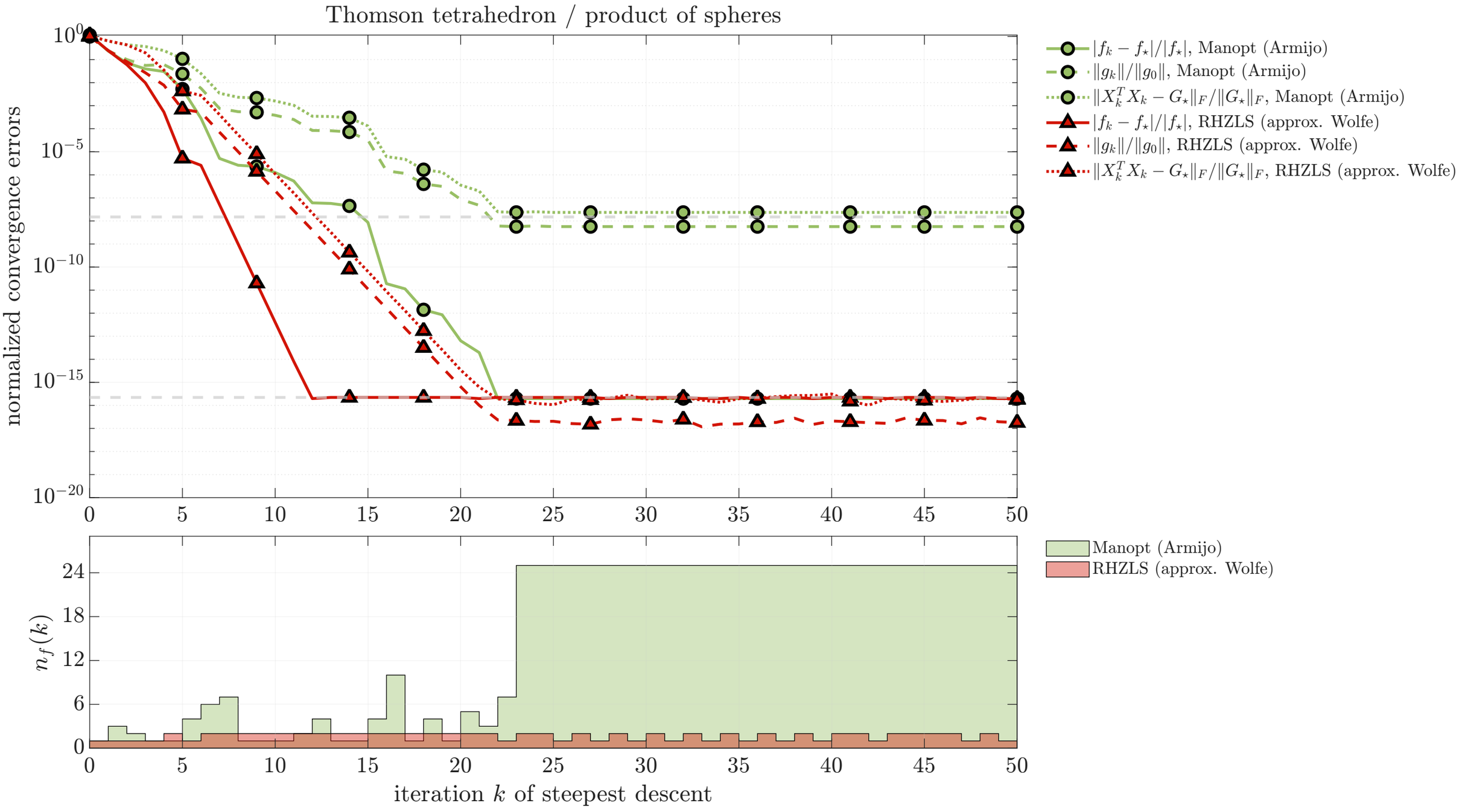

Thomson tetrahedron / product of spheres
normalized convergence errors
10^{0}
10^{-5}
10^{-10}
10^{-15}
10^{-20}
0
5
10
15
20
25
30
35
40
45
50
$|f_k - f_\star|/|f_\star|$, Manopt (Armijo)
$\|g_k\|/\|g_0\|$, Manopt (Armijo)
$\|X_k^T X_k - G_\star\|_F/\|G_\star\|_F$, Manopt (Armijo)
$|f_k - f_\star|/|f_\star|$, RHZLS (approx. Wolfe)
$\|g_k\|/\|g_0\|$, RHZLS (approx. Wolfe)
$\|X_k^T X_k - G_\star\|_F/\|G_\star\|_F$, RHZLS (approx. Wolfe)
$n_f(k)$
24
18
12
6
0
Manopt (Armijo)
RHZLS (approx. Wolfe)
iteration k of steepest descent